



CS105.3 - Database Management Systems

26.1 - Lab Sheet 02

Question No 01.

Suppose you are designing a database to store information about **Movies and Actors**. Each movie has a unique Movie ID, title, release year, and genre. Each actor has a unique Actor ID, name, and age. A movie can have **many actors**, and each actor can act in **one or more movies**. For each actor in a movie, the system should record the **specific role (character name)** played by the actor. Each movie must have **at least one actor**, and each actor must be associated with **at least one movie**.

Tasks:

1. Identify all the **entities** and list their **attributes** correctly.
2. Identify the **relationships** among the entities.
3. Specify the **cardinalities and participation constraints** for each relationship.
4. Draw an appropriate **ER diagram** for the given scenario.
5. Clearly state any **assumptions** you make.

Question No 02.

Scenario: Hospital Management System You are tasked with designing a database for a hospital that manages patient records, doctors, appointments, prescriptions, and medical tests. The hospital has the following requirements:

1. **Patient:** Each patient has a unique PatientID, Name, Address, Phone, and DateOfBirth. A patient can book multiple appointments, but an appointment is always associated with one patient.
2. **Doctor:** Each doctor has a unique DoctorID, Name, Specialty, and Phone. A doctor can see many patients, and each patient can be seen by many doctors (many-to-many relationship between Patient and Doctor).
3. **Appointment:** Each appointment has an AppointmentID, Date, and Time. Each appointment is scheduled with exactly one doctor and one patient.
4. **Prescription:** Each appointment can result in one or more prescriptions. A prescription has a unique PrescriptionID, DateIssued, and Dosage.
5. **Medication:** Medications prescribed to patients have a unique MedicationID, Name, and Description. Each prescription can include multiple medications, and the same medication can be prescribed multiple times.
6. **Medical Test:** Each patient can undergo multiple medical tests. Each test has a unique TestID, TestName, and Result. Medical tests are linked to an appointment but can also be ordered without one.

1. Design the ER diagram for the hospital management system based on the above scenario.
2. Consider the role of total vs. partial participation in the relationships between Patient and Appointment and explain how it will affect the ER diagram and the underlying tables.

Question No 03.

A university wants to design a database for its **Hospital Management System**. Based on the following description, draw an ER diagram with all entities, attributes, relationships, and cardinalities.

- The hospital has **Doctors**, each identified by a DoctorID (PK), with attributes Name, Specialization, and PhoneNumber.
- The hospital has **Patients**, each identified by a PatientID (PK), with attributes Name, DateOfBirth, and BloodGroup.
- The hospital has **Departments**, each identified by a DeptID (PK), with attributes DeptName and Location.
- Each **Doctor** belongs to exactly one **Department**; a Department can have many Doctors. (*WORKS_IN*)
- A **Doctor** can treat many **Patients**, and a Patient can be treated by many Doctors. The relationship **TREATS** has a relationship attribute DiagnosisDate.
- **Prescriptions** are a **weak entity**, dependent on a **Patient**. Each Prescription has a partial key PrescriptionNo, and attributes Medication and Dosage. A Patient can have many Prescriptions, but each Prescription belongs to exactly one Patient. (*HAS*)

Tasks:

1. Identify all strong entities and their primary keys.
2. Identify the weak entity and its partial key.
3. Show the relationship attribute on the appropriate relationship.
4. Indicate all cardinality ratios (1:1, 1:N, M:N).